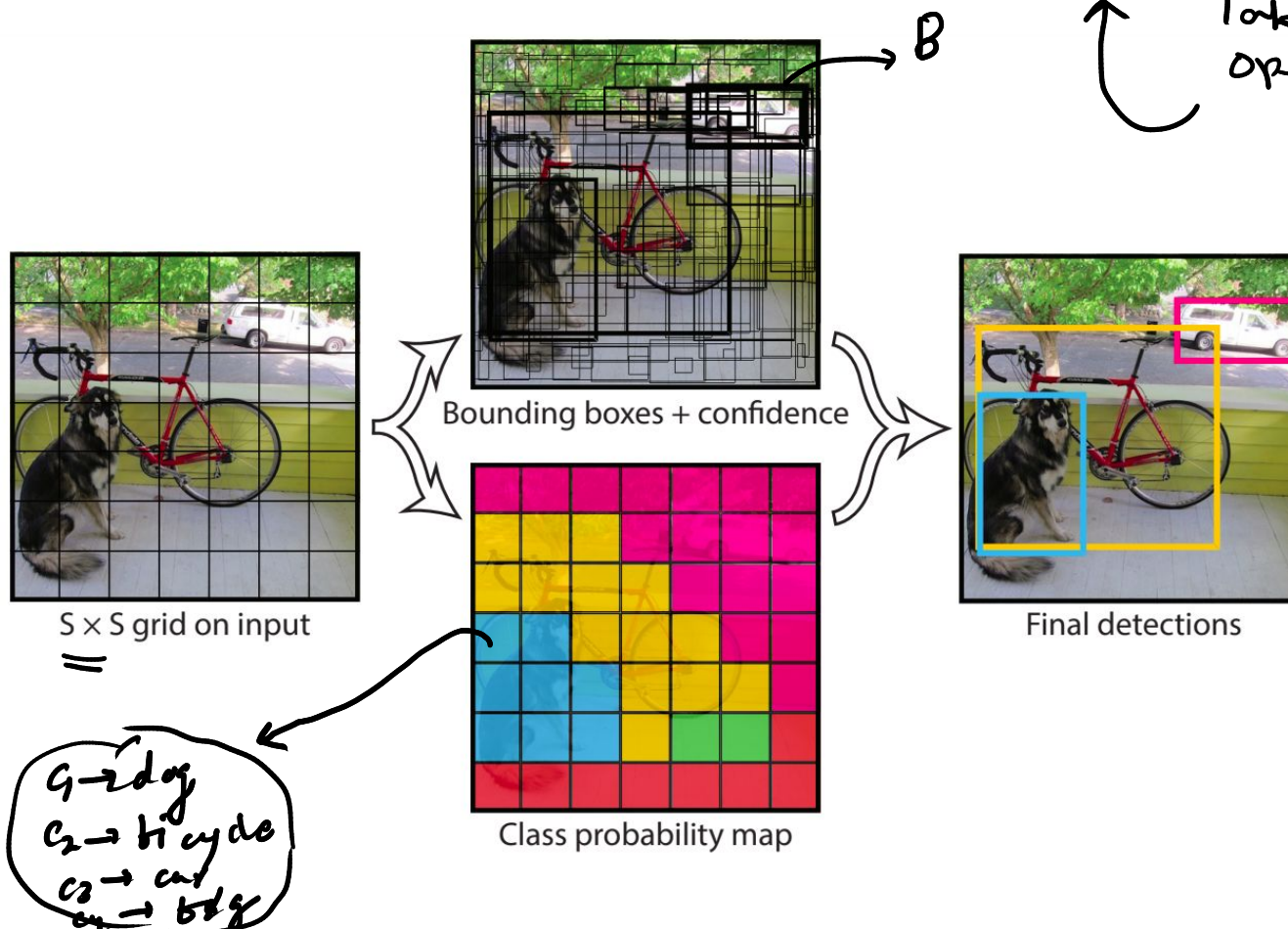


Region-free approach: YOLO



for each grid

Take Multiple BB
optimize over each grid



REF: You Only Look Once: Unified, Real-Time Object Detection, <https://arxiv.org/pdf/1506.02640.pdf>

$$[c_1, c_2, c_3, c_4]$$

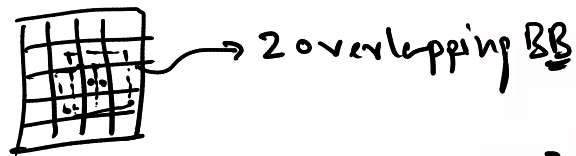
"RP", "YOLO"

→ Step 1: Divide the input image into a $G \times G$ grid.

• Step 2: For each grid cell, run a CNN that predicts y of the following form:

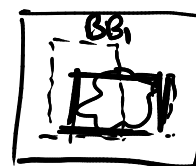
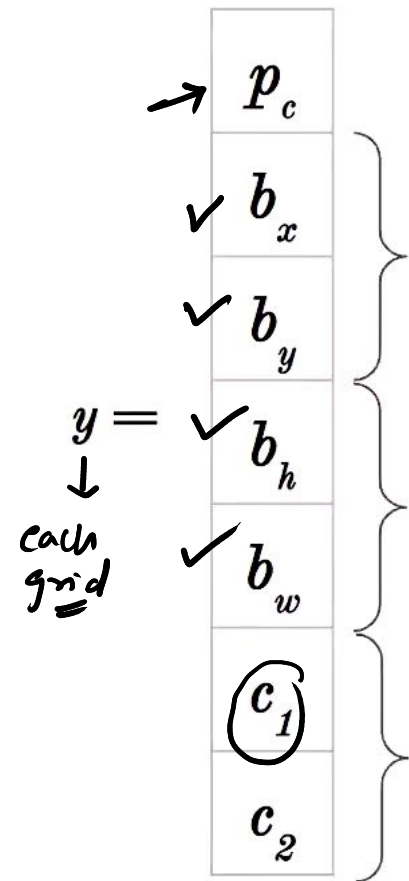
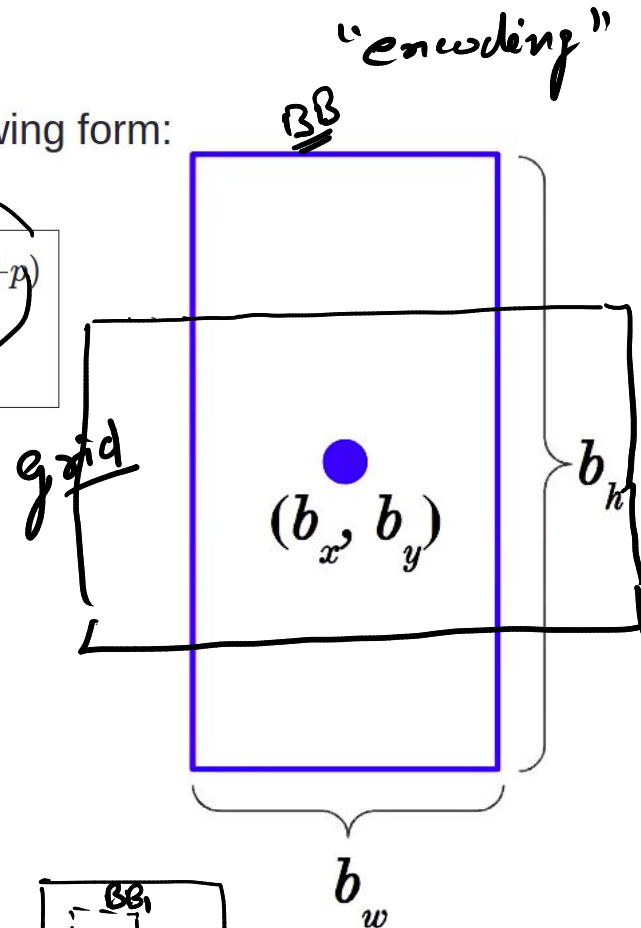
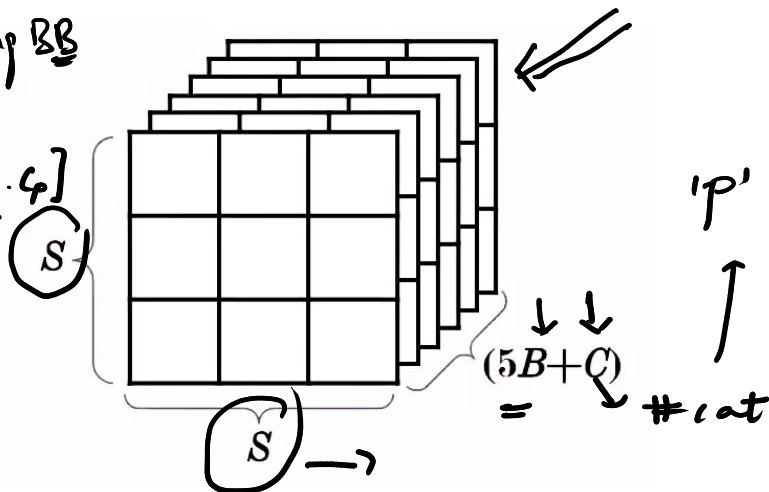
$$y = [p_c, b_x, b_y, b_h, b_w, \underbrace{c_1, c_2, \dots, c_p}_{\text{repeated } k \text{ times}}, \dots]^T \in \mathbb{R}^{G \times G \times k \times (5+p)}$$

• Step 3: Run the non-max suppression algorithm



$$y = [p_c, b_x, \dots, c_p, p_c, \dots, c_p]$$

$$\square \rightarrow [1, 0, 0, 0]$$



$$BB_1 \rightarrow P = 75\% \\ BB_2 \rightarrow P = 95\%$$